HERIOT-WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

DEPARTMENT OF ACTUARIAL MATHEMATICS AND STATISTICS

F78AA1

ACTUARIAL AND FINANCIAL MATHEMATICS A

16 December 2011

Time allowed 2 hours

Attempt all 6 questions

Total Marks 60

Only approved calculators may be used

1. At a rate of interest of 12% per annum effective, calculate:

   (i) the nominal rate of interest per annum compounded half-yearly;

   (ii) the effective rate of discount per half-year;

   (iii) the present value of £100 payable every half-year in advance for perpetuity.

   [4 marks]

2. A university student opens a bank account to save for a holiday which she plans to take when she graduates in three years' time. The payments she makes into the bank account are as follows:

   In the first year: £500 every half-year in advance;

   In the second year: £200 every quarter-year in advance;

   In the third year: £100 every month in arrears.

   Assume the bank account earns interest at a rate of 3% per annum effective (compound).

   (a) Calculate the value of her bank account at the end of the third year.

   [9 marks]

   (b) Calculate the present value at time 0 of the payments.

   [1 mark]

   (c) If interest was simple, rather than compound, would your answer to part (a) be higher or lower? Why?

   [2 marks]

   [Total: 12 marks]

continued ...

3. (a) Consider an annuity payable every time unit in arrears for n time units. The first payment is £1. Payments increase at a rate of 100j% per time unit effective.

Show, from first principles, that the present value at time 0 of these payments, at a rate of interest of i per time unit effective is:

$$\frac{1}{1+j}\, a_{\overline{n}|}^{i'}$$

at an appropriate rate of interest, i', which you should define.

[6 marks]

(b) A man takes out a loan of £125,000 to buy a supercar. He makes level monthly repayments in arrears over 20 years. The interest rate on the loan is 6% per annum effective. Calculate the amount of the level monthly repayment.

[4 marks]

(c) Immediately after the 120th repayment, due to a change in his financial situation, the man decides that he would like to alter the terms of the loan by changing the future repayments. For the remaining duration of the loan, he will make monthly repayments in arrears which will increase at a rate of 0.3% per month.

(i) Calculate the capital outstanding immediately after the 120th repayment.

(ii) Hence, calculate the amount of the first repayment he makes after this alteration occurs.

[7 marks]

[Total 17 marks]

4. The government issues a fixed interest bond on 1 January 2011 which pays half-yearly coupons at a rate of 7% per annum. The bond has a term of 10 years and will be redeemed at 106%.

An investor, who is subject to income tax at 20% and capital gains tax at 30%, purchases the bond at issue. The investor achieves a net monetary redemption yield of 7.635% per annum effective.

(a) Calculate the price (per £100 nominal) paid by the investor for the bond.

[6 marks]

(b) If the inflation rate is assumed to remain constant at a rate of 3% per annum, calculate the net real redemption yield the investor achieves on the bond.

[1 mark]

[Total: 7 marks]

continued ...

5. (a) Explain in words what is meant by the discounted payback period for a project.

[1 mark]

(b) An investment manager is considering investing in a project.

The project involves an initial investment of £1,000,000 on 1 January 2012 and will return cashflows of £100,000 on 1 January 2022, £200,000 on 1 January 2023, £300,000 on 1 January 2024, and so on, until a final cashflow of £1,000,000 is returned on 1 January 2031.

Assume a rate of interest of 7% per annum effective.

By considering the cashflows returned to the investor as an increasing annuity, show that the discounted payback period of the project is 7 years after the initial investment is made.

[7 marks]

(c) Explain why discounted payback period alone may not be considered a suitable measure for appraising an investment. State the name of a more suitable measure.

[2 marks]

[Total: 10 marks]

6. (a) State the main disadvantage of using Money Weighted Rate of Return compared to Time Weighted Rate of Return in order to measure the performance of an investment manager.

[1 mark]

(b) An investment fund had a fund value of £157 million on 31 December 2007, and £301 million on 31 December 2010. On 31 December 2008, the fund received a cash inflow of £100 million.

(i) Using linear interpolation as necessary, calculate an approximation for the Money Weighted Rate of Return for the period between 31 December 2007 and 31 December 2010.

(ii) The Money Weighted Rate of Return and the Time Weighted Rate of Return for the period between 31 December 2007 and 31 December 2010 are equal (when expressed as a percentage to two decimal places).
Using your approximation from part (b)(i), calculate the value of the fund on 31 December 2008 immediately before the cash inflow.

[9 marks]

[Total: 10 marks]

Total 60 marks